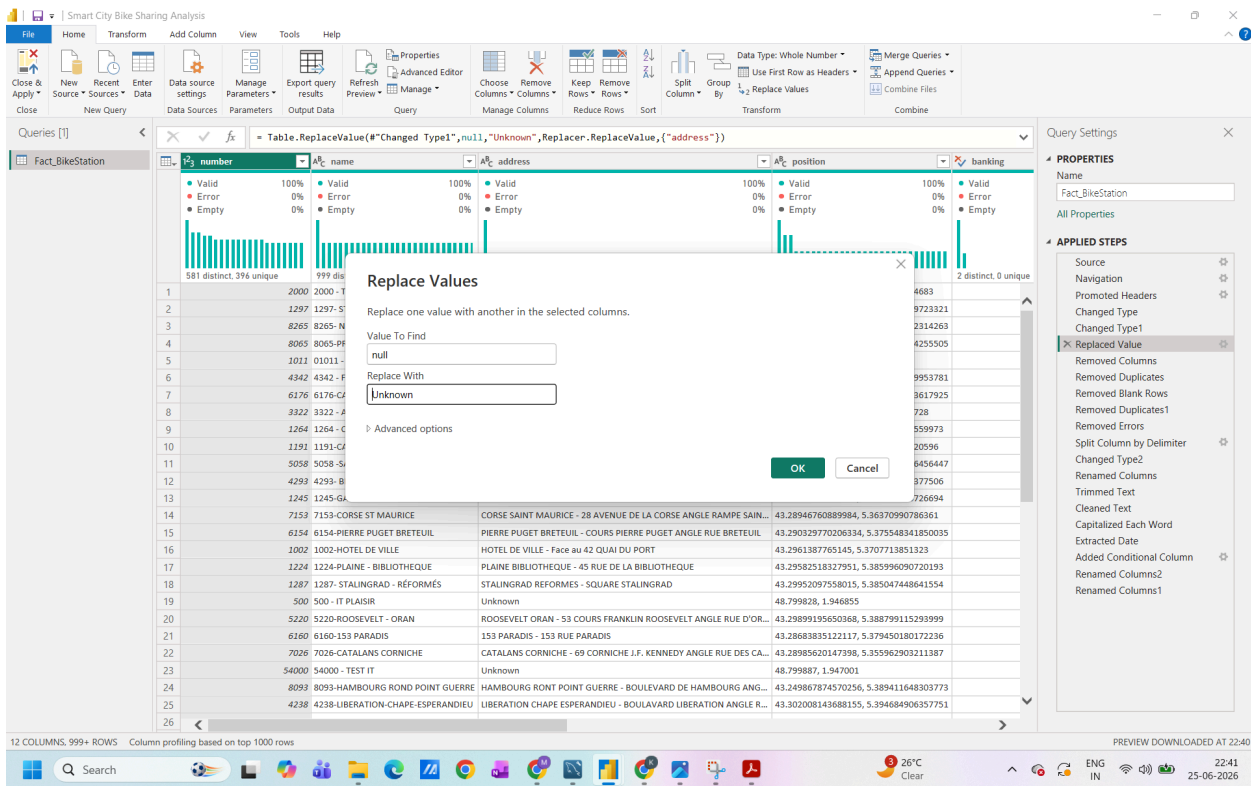


SMART CITY BIKE SHARING ANALYTICS

Screenshot : Data Cleaning & Transformation (Power Query)



Smart City Bike Sharing Analysis

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns Keep Remove Rows Sort Split Column Group By Data Type: Whole Number Merge Queries Append Queries Combine Files Combine

Queries [1] Fact_BikeStation

Table.SplitColumn(#[Removed Errors], "position", Splitter.SplitTextByDelimiter(",", QuoteStyle.None), {"position.1", "position.2"})

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

Comma

Split at

Left-most delimiter

Right-most delimiter

Each occurrence of the delimiter

Advanced options

Quote Character

None

Split using special characters

Insert special character

OK Cancel

Query Settings

PROPERTIES

Name

Fact_BikeStation

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Changed Type1

Replaced Value

Removed Columns

Removed Duplicates

Removed Blank Rows

Removed Duplicates1

Removed Errors

Split Column by Delimiter

Changed Type2

Renamed Columns

Trimmed Text

Cleaned Text

Capitalized Each Word

Extracted Date

Added Conditional Column

Renamed Columns2

Renamed Columns1

11 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Smart City Bike Sharing Analysis

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Queries [1] Fact_BikeStation

Table.TransformColumns(#[Renamed Columns],({"address", Text.Trim, type text}))

1.2 Latitude

1.2 Longitude

Trimmed Text

Query Settings

PROPERTIES

Name

Fact_BikeStation

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Changed Type1

Replaced Value

Removed Columns

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Queries [1]

Fact_BikeStation

Table.TransformColumns("Capitalized Each Word",{"Last Update", DateTime.Date, type date})

status Contract Name Bike Stands Available Bike Stands Available Bikes Last Update

100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid

0% Error 0% Error 0% Error 0% Error 0% Error 0% Error

0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty

2 distinct, 0 unique 23 distinct, 8 unique 44 distinct, 6 unique 37 distinct, 3 unique 40 distinct, 7 unique 148 distinct, 100 unique

1	33502653	Open	Maribor	1	0	0	02-06-2022
2	3784238	Open	Marseille	9	0	0	06-04-2022
3	39387542	Open	Marseille	15	0	0	27-09-2022
4	72088524	Open	Marseille	15	15	0	08-06-2022
5	-1.5575	Open	Nantes	0	0	0	23-05-2022
6	39367826	Open	Marseille	10	0	0	13-09-2022
7	35593764	Open	Marseille	24	24	0	06-09-2022
8	38741804	Open	Marseille	20	13	7	14-09-2022
9	31132216	Open	Marseille	10	0	0	12-09-2022
10	76930662	Open	Marseille	20	2	0	03-10-2022
11	31285286	Open	Marseille	10	10	0	13-09-2022
12	36000803	Open	Marseille	14	0	0	26-09-2022
13	31500047	Open	Marseille	15	0	0	03-10-2022
14	33709908	Open	Marseille	10	0	0	10-10-2022
15	75548342	Open	Marseille	8	0	0	26-09-2022
16	70771385	Open	Marseille	32	0	0	17-10-2022
17	35996091	Open	Marseille	16	0	0	24-10-2022
18	35047449	Open	Marseille	17	0	0	04-10-2022
19	1.946855	Open	Jcdcauxbike	1	0	1	30-08-2022
20	38799115	Open	Marseille	10	0	0	08-11-2022
21	37945018	Open	Marseille	14	0	0	22-11-2022
22	35962903	Open	Marseille	17	0	0	11-10-2022
23	1.947001	Open	Nancy	1	1	0	28-11-2022
24	39411648	Open	Marseille	15	0	0	22-11-2022
25	34684906	Open	Marseille	12	0	0	24-10-2022
26							

11 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Smart City Bike Sharing Analysis

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [1]

Fact_BikeStation

Table.AddColumn("Extracted Date", "Open Station Flag", each if [status] = "Open" then 1 else 0)

status Contract Name Bike Stands Available Bike Stands Available Bikes Last Update

100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid

0% Error 0% Error 0% Error 0% Error 0% Error 0% Error

0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty

2 distinct, 0 unique 23 distinct, 8 unique 44 distinct, 6 unique 37 distinct, 3 unique 40 distinct, 7 unique 148 distinct, 100 unique

1	33502653	Open	Maribor	1	0	0	02-06-2022
2	3784238	Open	Marseille	9	0	0	06-04-2022
3	39387542	Open	Marseille	15	0	0	27-09-2022
4	72088524	Open	Marseille	15	15	0	08-06-2022
5	-1.5575	Open	Nantes	0	0	0	23-05-2022
6	39367826	Open	Marseille	10	0	0	13-09-2022
7	35593764	Open	Marseille	24	24	0	06-09-2022
8	38741804	Open	Marseille	20	13	7	14-09-2022
9	31132216	Open	Marseille	10	0	0	12-09-2022
10	76930662	Open	Marseille	20	2	0	03-10-2022
11	31285286	Open	Marseille	10	10	0	13-09-2022
12	36000803	Open	Marseille	14	0	0	26-09-2022
13	31500047	Open	Marseille	15	0	0	03-10-2022
14	33709908	Open	Marseille	10	0	0	10-10-2022
15	75548342	Open	Marseille	8	0	0	26-09-2022
16	70771385	Open	Marseille	32	0	0	17-10-2022
17	35996091	Open	Marseille	16	0	0	24-10-2022
18	35047449	Open	Marseille	17	0	0	04-10-2022
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23	1.947001	Open	Nancy	1	1	0	28-11-2022
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25	34684906	Open	Marseille	12	0	0	24-10-2022
26							

12 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Smart City Bike Sharing Analysis

FileHomeTransformAdd ColumnViewToolsHelp

Close & ApplyNew SourceRecent SourcesEnter DataData source settingsManage ParametersExport query resultsRefresh PreviewQuery

Choose Remove ColumnsKeep RowsRemove RowsSplit ColumnGroup By

PropertiesAdvanced EditorManage ColumnsManage RowsReduce RowsSort

Data Type: Whole NumberUse First Row as HeadersReplace Values

Merge QueriesAppend QueriesCombine FilesCombine

Table.RenameColumns(#"Renamed Columns2",{"number","Station ID"},{"name","Station Name"},{"address","Address"},{"Contract Name"},

Fact_BikeStation

Station IDStation NameAddressLatitudeLongitude

Valid100%Valid100%Valid100%Valid100%Valid100%

Error0%Error0%Error0%Error0%Error0%

Empty0%Empty0%Empty0%Empty0%Empty0%

120002000 - TEST DSI PLAISIRUnknown46.5488766815.635026

212971297- ST FÉRÉOL DAVSO ST FERREOL DAVSO - RUE FRANCOIS DAVSO ANGLE RUE SAINT FERREOL43.293485745.37842

382658265- NÉGRESKO - PAULETNEGRESKO PAULET - FACE AU 42 RUE NEGRESKO43.269318565.3893875

480658065-PROMENADE POMPIDOU PALM PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...43.267272525.3720885

5101101011 - BORNE TEST NANTES 1Borne test 147.19504-1.55

643424342 - FLAMMARION MONTRICHETFLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...43.305993455.393678

761766176-CANTINI ROUET CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI43.284775985.3855937

833223322 - ARENC CHANTERAC32 rue de Chanterac 13003 Marseille43.312337035.3687418

912641264 - CANEBIERE DUGOMMIERCANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGO...43.297896315.3811322

1011911191-CANEBIERE ST FERREOLCANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL43.296207675.3769906

1150585058 - SAKAKINI - SAINT PIERRESAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIE...43.292800865.4012852

1242934293- BLANCARDEBLANCARDE - GARE DE LA BLANCARDE43.296263415.4060008

1312451245-GAMBETTA - LAFAYETTEGAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA43.299027945.3815900

1471537153-CORSE ST MAURICECORSE SAINT MAURICE - 28 AVENUE DE LA CORSE ANGLE RAMPE SAIN...43.289467615.3637095

1561546154-PIERRE PUGET BRETEUILPIERRE PUGET BRETEUIL - COURS PIERRE PUGET ANGLE RUE BRETEUIL43.290329775.3755483

1610021002-HOTEL DE VILLEHOTEL DE VILLE - Face au 42 QUAI DU PORT43.296138785.3707713

1712241224-PLAINE - BIBLIOTHEQUEPLAINE BIBLIOTHEQUE - 45 RUE DE LA BIBLIOTHEQUE43.295825185.3859960

1812871287- STALINGRAD - RÉFORMÉSSTALINGRAD REFORMES - SQUARE STALINGRAD43.299520985.3850474

19500500 - IT PLAISIRUnknown48.7998281.9468

2052205220-ROOSEVELT - ORANROOSEVELT ORAN - 53 COURS FRANKLIN ROOSEVELT ANGLE RUE D'OR...43.28991965.3887991

2161606160-153 PARADIS153 PARADIS - 153 RUE PARADIS43.286838355.379450

2270267026-CATALANS CORNICHECATALANS CORNICHE - 69 CORNICHE J.F. KENNEDY ANGLE RUE DES CA...43.28985625.3559625

235400054000 - TEST ITUnknown48.7998871.9470

2480938093-HAMBOURG ROND POINT GUERREHAMBOURG RONT POINT GUERRE - BOULEVARD DE HAMBOURG ANG...43.249867875.3894116

2542384238-LIBERATION-CHAPE-ESPERANDIEULIBERATION CHAPE ESPERANDIEU - BOULAVARD LIBERATION ANGLE R...43.302008145.3946849

26

12 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Search

Very humid Now

ENG IN

22:50

25-06-2026

Query Settings

PROPERTIES

Fact_BikeStation

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Changed Type1

Replaced Value

Removed Columns

Removed Duplicates

Removed Blank Rows

Removed Duplicates1

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Smart City Bike Sharing Analysis

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Choose Remove ColumnsKeep RowsRemove RowsSplit ColumnGroup By

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Merge QueriesAppend QueriesCombine FilesCombine

Table.RenameColumns(#"Renamed Columns2",{"number","Station ID"},{"name","Station Name"},{"address","Address"},{"Contract Name"},

Fact_BikeStation

statusCityBike StandsEmpty DocksAvailable BikesLast UpdateOpen Stati

Valid100%Valid100%Valid100%Valid100%Valid100%Valid100%

Error0%Error0%Error0%Error0%Error0%Error0%

Empty0%Empty0%Empty0%Empty0%Empty0%Empty0%

122 distinct, 0 unique

25 distinct, 8 unique

44 distinct, 1 unique

37 distinct, 3 unique

40 distinct, 7 unique

148 distinct, 100 unique

1OpenMarilbor102-06-2022

2OpenMarseille90-06-04-2022

3OpenMarseille150-27-09-2022

4OpenMarseille150-08-06-2022

5OpenNantes00-23-05-2022

6OpenMarseille100-13-09-2022

7OpenMarseille240-06-09-2022

8OpenMarseille2013-14-09-2022

9OpenMarseille100-12-09-2022

10OpenMarseille202-03-10-2022

11OpenMarseille1010-13-09-2022

12OpenMarseille140-26-09-2022

13OpenMarseille150-03-10-2022

14OpenMarseille100-10-10-2022

15OpenMarseille80-26-09-2022

16OpenMarseille320-17-10-2022

17OpenMarseille160-24-10-2022

18OpenMarseille170-04-10-2022

19OpenJcdcauxbike11-30-08-2022

20OpenMarseille100-08-11-2022

21OpenMarseille140-22-11-2022

22OpenMarseille170-11-10-2022

23OpenNancy11-28-11-2022

24OpenMarseille150-22-11-2022

25OpenMarseille120-24-10-2022

26

12 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Search

Very humid Now

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22:50

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Renamed Columns

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Cleaned Text

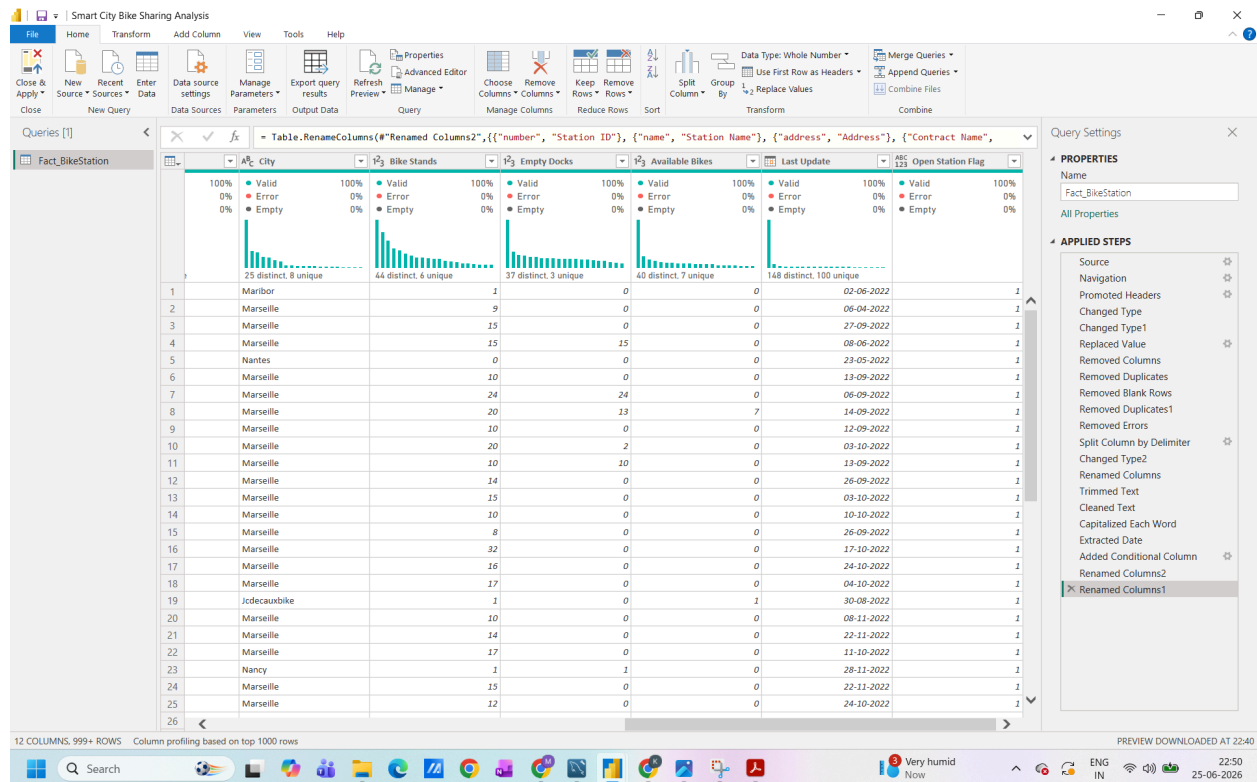
Capitalized Each Word

Extracted Date

Added Conditional Column

Renamed Columns2

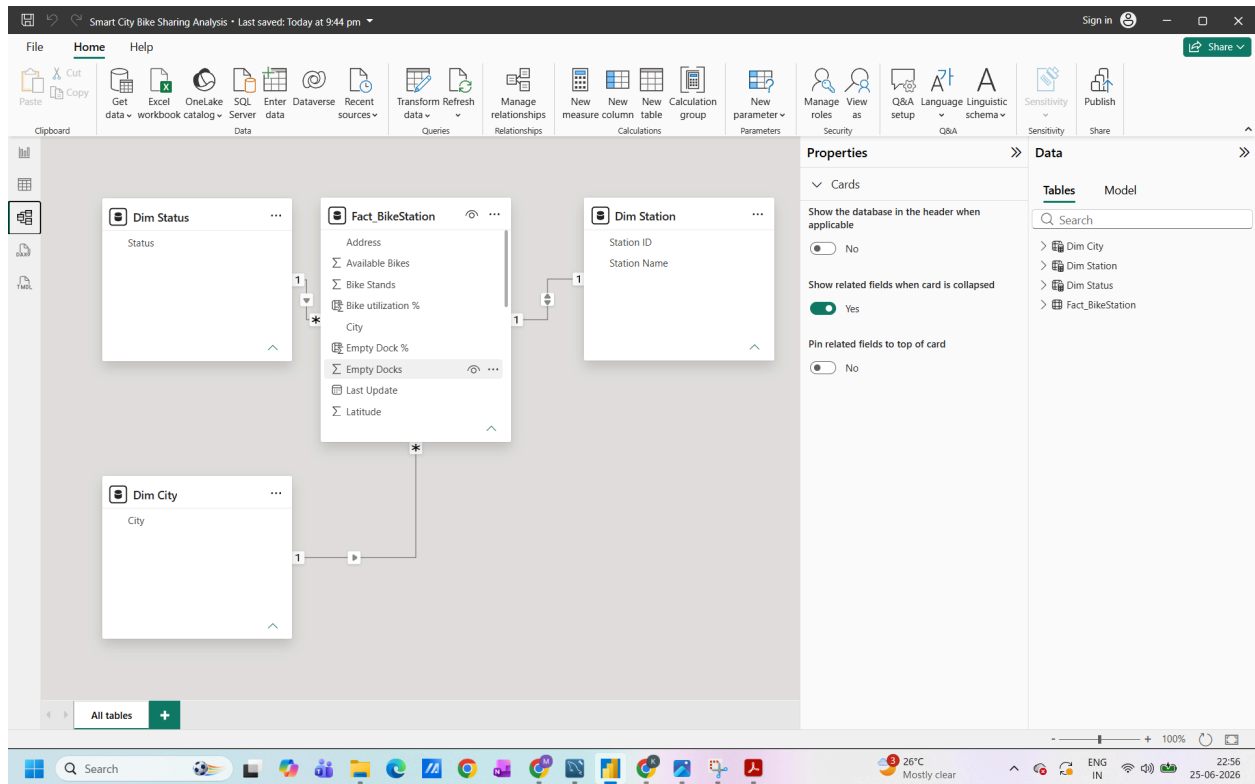
Renamed Columns1



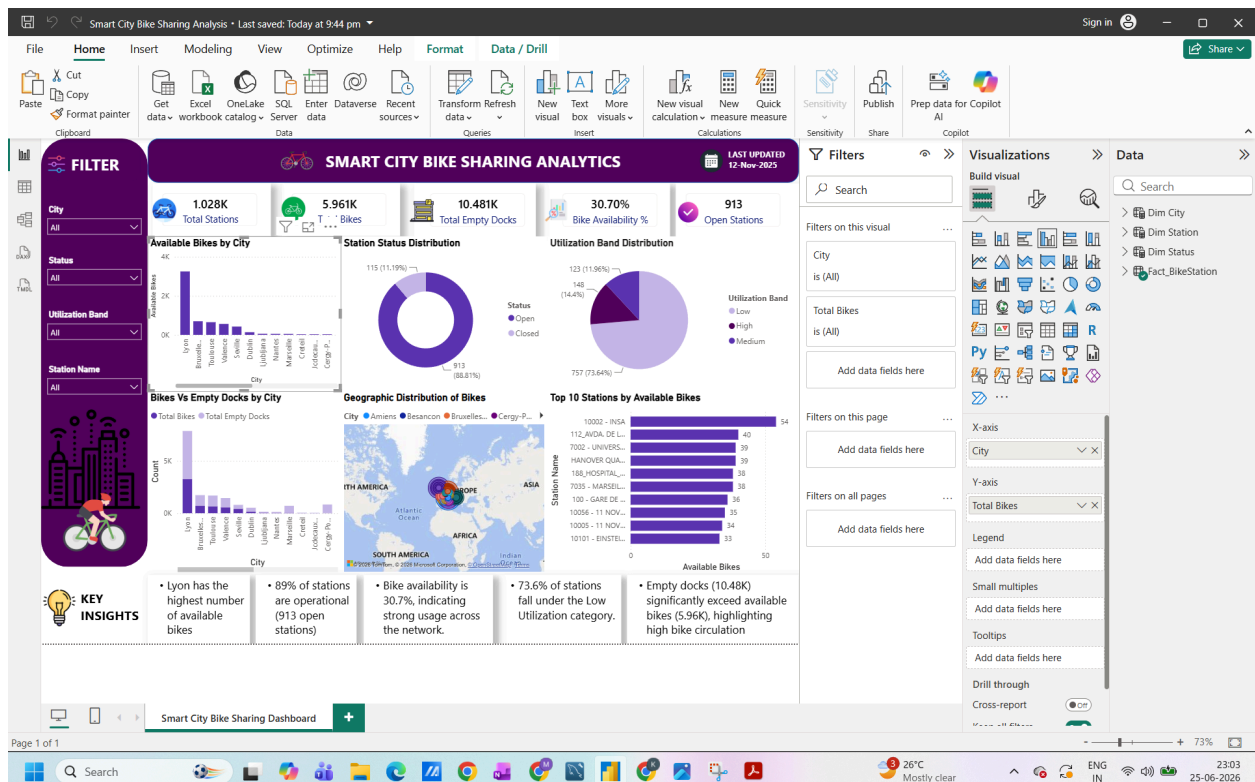
Data was cleaned using Power Query by removing null values, fixing data types,split column by delimiter,added conditional column and renaming columns for better readability. Unnecessary columns were removed to improve model performance.

Screenshot : Data Model View

A star schema model was created by connecting fact and dimension tables using City,Station and Status keys.



Screenshot Chart Details (Power BI Dashboard)



Available Bikes by City (Bar Chart)

- This chart shows the distribution of available bikes across different cities.
- It helps identify cities with high and low bike availability.
- Useful for understanding demand and supply imbalance in each city.

Station Status Distribution (Donut Chart)

- This chart displays the proportion of Open vs Closed stations.
- It shows the operational efficiency of the bike-sharing system.
- Most stations are operational, indicating a stable system.

Utilization Band Distribution (Pie Chart)

- This chart groups stations into Low, Medium, and High utilization bands.
- It helps understand how effectively bikes are being used.
- Most stations fall under the low utilization category.

Bikes vs Empty Docks by City (Bar Chart)

- This compares available bikes and empty docking spaces in each city.
- It highlights bike circulation and station capacity usage.
- Empty docks are higher than available bikes, showing active movement.

Geographic Distribution of Bikes (Map Visual)

- This map shows the location of bike stations using latitude and longitude.
- Bubble size indicates bike availability in each region.
- It helps visualize spatial distribution of bike-sharing stations.

Top 10 Stations by Available Bikes (Bar Chart)

- This chart shows stations with the highest number of available bikes.
- It helps identify high-supply stations in the network.
- Some stations show significantly higher availability than others.